

TEGYULTYA, Russian Federation

WMO: 249670

Lat: 60.467N

Lon: 130.000E

Elev: 171

StdP: 99.28

Time Zone: 9.00 (E09)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-48.7	-47.1	-50.5	0.0	-48.5	-48.9	0.0	-47.0	4.3	-27.2	3.7	-26.9	0.0	270	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.5	30.3	19.5	28.2	18.4	26.2	17.5	21.1	27.5	19.8	25.7	18.5	24.2	1.9	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.1	14.1	23.9	17.7	12.9	22.9	16.3	11.9	21.4	62.2	27.9	57.8	25.8	53.4	24.5	30.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.6	4.4	3.7	DB	-50.7	33.2	1.7	2.5	-52.0	35.0	-53.0	36.5	-53.9	37.9	-55.2	39.7	(4)
(5)			WB	-50.7	24.5	1.6	2.9	-51.8	26.7	-52.8	28.4	-53.7	30.0	-54.9	32.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-8.3	-35.9	-32.6	-19.4	-5.0	7.6	14.9	17.8	14.5	5.2	-6.6	-26.0	-35.5	(6)	
	DBStd	20.64	6.36	6.30	7.36	6.59	3.92	3.48	3.50	3.78	4.21	6.30	8.12	6.43	(7)	
	HDD10.0	7238	1422	1192	912	451	94	4	1	7	152	514	1080	1411	(8)	
	HDD18.3	9764	1681	1426	1170	701	332	112	51	129	393	772	1330	1669	(9)	
	CDD10.0	569	0	0	0	0	21	150	244	147	8	0	0	0	(10)	
	CDD18.3	54	0	0	0	0	0	8	36	11	0	0	0	0	(11)	
	CDH23.3	1161	0	0	0	0	16	292	622	228	3	0	0	0	(12)	
	CDH26.7	366	0	0	0	0	2	72	224	68	0	0	0	0	(13)	
(14) Wind	WSAvg	1.2	0.6	0.6	1.1	1.5	1.9	1.5	1.5	1.4	1.5	1.4	1.0	0.7	(14)	
(15) Precipitation	PrecAvg	293	13	9	7	9	27	34	48	52	40	22	19	12	(15)	
	PrecMax	438	30	20	18	34	71	80	97	127	95	65	66	23	(16)	
	PrecMin	193	3	2	2	0	6	8	11	8	12	7	5	4	(17)	
	PrecStd	67	7	4	4	8	14	19	27	31	21	13	13	4	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-14.7	-11.1	6.4	15.2	26.1	30.7	35.3	31.7	22.5	8.0	0.8	-13.7	(19)	
		MCWB	-15.1	-11.6	2.1	6.3	14.5	18.2	21.9	20.1	14.3	5.4	-0.6	-14.0	(20)	
	2%	DB	-19.2	-15.3	1.7	11.6	22.5	28.8	31.6	28.4	18.5	5.4	-4.5	-19.4	(21)	
		MCWB	-19.5	-15.8	-1.8	4.8	11.5	17.3	20.7	19.0	12.2	2.8	-5.4	-19.7	(22)	
	5%	DB	-23.0	-18.9	-0.9	9.1	20.2	26.8	29.1	25.9	15.8	3.4	-9.6	-22.0	(23)	
		MCWB	-23.2	-19.3	-3.7	3.1	10.6	16.7	19.6	17.8	10.5	1.5	-10.2	-22.2	(24)	
	10%	DB	-26.3	-21.9	-4.6	6.6	17.4	24.7	27.0	23.4	13.4	1.9	-14.1	-25.2	(25)	
		MCWB	-26.5	-22.2	-6.7	1.8	9.2	15.8	18.7	16.4	9.1	0.6	-14.5	-25.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-15.3	-11.7	2.1	7.4	15.4	20.5	24.3	22.2	14.9	5.6	-0.4	-13.9	(27)	
		MCDB	-15.0	-11.4	5.8	13.7	25.3	27.1	31.7	28.2	21.1	7.7	0.3	-13.8	(28)	
	2%	WB	-19.5	-15.8	-1.4	5.0	12.5	18.8	21.8	20.3	13.3	3.4	-5.4	-19.7	(29)	
		MCDB	-19.2	-15.4	1.5	11.0	19.8	25.9	29.1	26.1	17.8	4.7	-4.6	-19.4	(30)	
	5%	WB	-23.2	-19.3	-3.7	3.5	11.1	17.6	20.7	18.8	11.4	1.9	-10.2	-22.2	(31)	
		MCDB	-23.0	-18.9	-1.0	8.6	18.0	24.6	26.9	24.0	14.8	2.8	-9.6	-22.0	(32)	
	10%	WB	-26.5	-22.2	-6.5	2.0	9.9	16.5	19.5	17.2	9.6	0.5	-14.5	-25.3	(33)	
		MCDB	-26.3	-21.9	-4.5	6.3	15.9	23.4	25.2	22.6	12.6	2.0	-14.1	-25.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.3	14.9	20.4	17.9	14.0	15.2	14.5	14.4	12.9	9.6	10.8	7.9	(35)	
		MCDBR	12.5	17.6	22.3	18.3	20.4	20.3	20.1	19.7	17.4	10.0	14.3	11.4	(36)	
		MCWBR	12.3	17.0	18.9	11.7	10.6	10.1	10.4	10.6	11.1	7.5	13.6	11.2	(37)	
	5% WB	MCDBR	12.5	17.6	21.4	17.3	16.6	17.4	16.4	16.7	14.9	8.2	13.8	11.4	(38)	
		MCWBR	12.3	17.0	18.2	11.5	9.4	9.3	9.3	9.4	10.0	6.6	13.2	11.2	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.232	0.271	0.309	0.376	0.453	0.427	0.509	0.460	0.359	0.298	0.239	0.205	(40)	
	taud		1.986	2.047	2.041	2.002	2.030	2.232	1.981	2.090	2.347	2.287	2.016	1.912	(41)	
	Ebn at Noon		504	703	797	805	767	803	712	717	753	681	482	369	(42)	
	Edh at Noon		45	79	112	143	151	126	158	131	85	63	43	30	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.49	1.54	3.26	5.11	5.21	5.85	5.35	4.20	2.65	1.33	0.66	0.26	(44)	
	RadStd		0.03	0.06	0.12	0.18	0.48	0.41	0.38	0.32	0.18	0.14	0.04	0.01	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	+0.70	+1.38	+1.29	N/A	N/A	N/A	-238	-240	N/A	N/A

Nomenclature: See separate page